

Unsupervised Learning of Global Demographic Regimes from Longitudinal Population Pyramid Structures

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ABSTRACT

Population pyramids provide a compact representation of the age structure of a population, yet comparative demographic analyses often reduce them to selected summary indicators or examine them at isolated time points. This study presents an unsupervised learning framework for analysing complete population pyramid structures across 201 geographical entities from 1950 to 2020. Using normalised age cohort proportions from the United Nations World Population Prospects, Principal Component Analysis was first used to obtain a compact representation of population structure. Two complementary analytical perspectives were then developed. The Global Demographic Evolution framework clustered countries at successive decadal time points to examine changes in the worldwide configuration of demographic regimes. The Country-Specific Demographic Evolution framework represented each country as a longitudinal trajectory through principal component space and compared these trajectories using functional smoothing, Dynamic Time Warping, and spectral clustering. The results show that global demographic change has not followed a single convergent pathway. Instead, youthful, transitional, post transitional ageing, and migration shaped regimes coexist across regions. The trajectory analysis further reveals distinct long term pathways that are partly consistent with the Demographic Transition Model, while also exposing variations in timing, pace, and structural form. This framework demonstrates the value of treating the population pyramid as an evolving demographic object.

Introduction

The population pyramid of a country is not only the primary graphical method that summarises the age-sex distribution of a given population at a given time point, but also a diagnostic tool for assessing a nation's developmental trajectory and future population trends by providing information about fertility, mortality, and migration^{1,2}. It is constructed by arranging successive age-sex cohorts typically in five-year intervals into two back-to-back horizontal bar graphs, with males conventionally displayed on the left and females on the right². These age sex structures are commonly classified into three broad types, namely expansive, stationary, and constrictive population pyramids, each of which reflects a distinct demographic regime associated with differing patterns of population growth and ageing³, as demonstrated in Figure 1.

The population pyramid serves as a multifaceted information provider encapsulating a nation's socio-economic status, policy history, and cultural evolution. The pyramid's morphology reveals levels of development and dependency ratios, i.e., whether a country's labour market can support its dependent populations and sustain long-term economic growth^{2,4}. The impact of government policies such as China's one-child policy, which restricted fertility in response to perceived overpopulation pressure, is reflected in the evolution of the country's population pyramid over time^{5,6}. The economic development in nations like South Korea has driven a cultural preference toward smaller families, given the high opportunity cost of parenting and gender-role imbalances^{7,8}. Net immigration patterns serve as a signal of a country's relative quality of life, as prosperity and robust labour markets attract external migrants. Conversely, instability and economic decline often repel them, causing cohort deficits⁹.

The depleting force of the population pyramid, mortality, reflects historical traumas such as wars that leave deep, lopsided gaps in the adult male sections^{10,11}, and global pandemics such as COVID-19 or localized outbreaks such as Ebola Virus Disease (EVD) epidemic in West Africa (2014–2016) that cause sudden structural shocks^{12,13}. Severe famines such as the

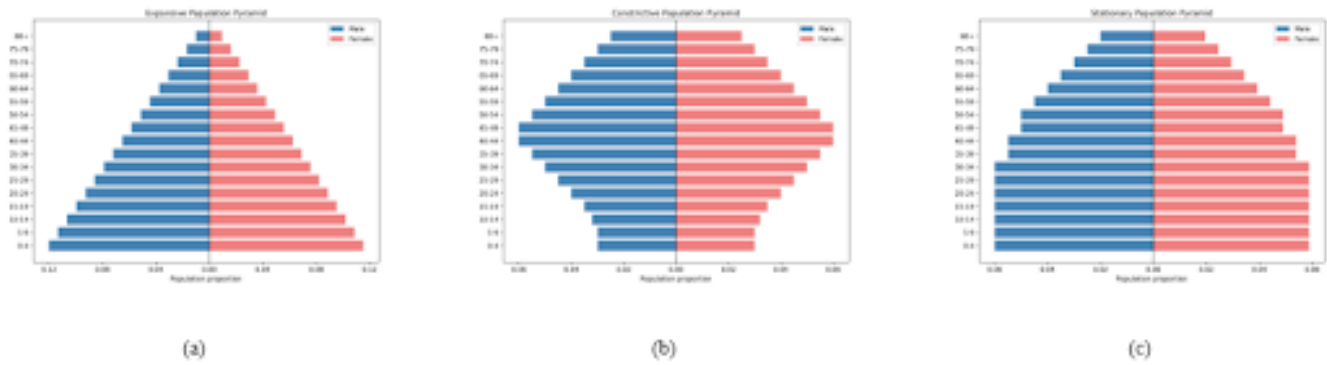


Figure 1. Representative population pyramid typologies illustrating (a) expansive, (b) constrictive, and (c) stationary age structures. The expansive form exhibits a broad base indicative of high fertility and a youthful population; the constrictive form reflects reduced birth rates and an ageing demographic profile; and the stationary form denotes a relatively balanced age distribution associated with stable population growth.

15 to 55 million deaths during China's Great Leap Forward (1959–1961) or the three million deaths during India's 1943 Bengal Famine resulted in missing birth cohorts and skewed dependency ratios^{14–16}. The prevalence of communicable and non-communicable diseases also affects the mortality reflected by the vertical slope of the population pyramid^{11,17}.

The Demographic Transition Model (DTM) visualizes how the developmental shifts fundamentally reshape a nation's population pyramid over time^{18,19}. At the first stage, in a society where traditional cultural norms prioritize large families despite low life expectancy, an expansive population pyramid with a wide base and concave sides is observed^{18,19}. As society undergoes developments such as improved sanitation and medical advancements, the age structure begins to broaden across the working-age bands, due to the maturation of young cohorts to adulthood without premature deaths^{19,20}. In the later stages, with an increase in urbanization and educational attainment, high opportunity costs of urban parenting cause the pyramid's base to contract, shifting the morphology from a triangle to a stationary or regressive structure^{21,22}. Although DTM is an ideal framework representing a linear evolution from traditional, non-industrial societies to modern, urban ones, real-world population pyramids rarely follow this path perfectly^{18,20}. Instead, they also reflect historical demographic signatures caused by sudden structural shocks such as wars, disease outbreaks, famine, and policy interventions¹¹.

Population pyramids inform multi-sectoral policy and health interventions^{2,23}. Quantifying the association between population structure, summarized by PoPDivergence, a scalar metric that condenses the information of a population pyramid into a single value, and cause-specific mortality rates, reveals which demographic profiles carry the highest risk for specific health hazards and enable reliable mortality predictions²³. Researchers forecast the demand for hospital services by analyzing the age-specific cohorts of population pyramids, where wide bases signal a need for pediatric and maternal services, while top-heavy structures with expanding senior cohorts show the need for long-term planning of geriatric care^{24,25}. Japan initiated "Society 5.0," leveraging AI and big data to address depopulation, informed by its aging population dynamics^{26,27}. Diffusion models have been designed to simulate how changes in population density across regions influence individual mobility, which is essential for planning urban traffic engineering and public health interventions during emergencies^{28,29}.

Current demographic research effectively categorizes countries using discrete indicators and cross-sectional pyramid shapes^{30–32}. Existing studies have successfully used hierarchical clustering and symbolic descriptions to group countries based on population pyramid shapes, such as expansive, constrictive, or stationary structures at specific time points^{30–32}. Research has used compositional data analysis and Aitchison distance to identify specific countries that epitomize the global population structure at different points in time between 1990 and 2080^{33,34}. While longitudinal studies exist for individual parameters like fertility or mortality convergence, the integrated morphological evolution of the complete pyramid remains under-researched in the absence of completely data-driven unsupervised learning applications to the high-dimensional trajectory of the entire age distribution throughout the post-WWII era^{35,36}.

The multidimensional nature of population pyramids gives rise to an analytical problem that is not easily addressed through isolated demographic indicators alone, namely, how broader structural regularities may be identified directly from the data themselves^{2,31}. Within the quantitative literature, approaches to this problem are commonly distinguished as supervised and unsupervised learning^{37,38}. In supervised learning, the analysis proceeds from labelled data, that is, from observations for which the outcome or class of interest is already known. The task is then to estimate a rule by which that predefined outcome may be inferred from the observed features^{37,38}. In this respect, supervised learning bears a close relationship to much of statistical modelling, particularly where one begins with a specified response variable and seeks to characterise or predict it

from a set of explanatory variables³⁹. Such an approach is therefore guided not only by the empirical data themselves, but also by prior analytical decisions concerning the response to be studied, the form of the labels, and the structure of the relationship to be learned^{37,39}.

Unsupervised learning differs in a fundamental respect. It does not begin with externally assigned labels, prespecified classes, or a predetermined response variable^{37,38}. Its purpose is instead to examine the internal organisation of the data and to identify regularities, affinities, and groupings that arise from the observations themselves⁴⁰. In clustering, this means that the analysis does not impose categories in advance, but seeks to determine which entities are empirically similar on the basis of their observed structure⁴⁰. The distinctive value of this approach lies in its genuinely data-driven character. Rather than asking whether the data conform to an already assumed classification, it allows broader configurations, latent relationships, and temporal tendencies to emerge from the empirical record itself^{37,38}. This is particularly important in sociological and demographic interpretation, where classifications based too strongly on prior expectation may introduce an element of intuitive bias on the part of the researcher. By avoiding the need to prescribe the relevant categories beforehand, unsupervised learning permits the data to speak more directly, allowing the principal patterns of resemblance and differentiation to be inferred before they are interpreted substantively⁴⁰. For demographic data, whose structure is often high-dimensional and whose evolution through time may not be adequately represented by a small number of prespecified indicators, such an approach is especially valuable^{2,31}. It permits the identification of meaningful patterns without requiring strong a priori presumptions regarding which populations should resemble one another or how their demographic trajectories ought to be ordered. Although unsupervised methods are not entirely free of methodological choices, they are distinguished by the fact that the principal structure to be interpreted is inferred from the data rather than prescribed in advance by the analyst^{37,40}.

Clustering is one of the principal techniques within unsupervised learning^{37,38,40}. Its purpose is to partition observations into groups in such a way that entities within the same group are more similar to one another, according to the features under study, than to entities in other groups^{40,41}. The resulting clusters are not predefined classes, but empirical groupings inferred from the structure of the data itself⁴⁰. In this way, clustering provides a systematic means of identifying underlying patterns, common structural characteristics, and broad developmental tendencies within complex datasets^{37,40}.

In the present context, a population pyramid based trajectory may be understood as the path traced by a country's age structure through a common analytical frame over the period from 1950 to 2020^{2,42}. It provides a quantitative representation of a fundamentally qualitative demographic journey, converting successive population pyramids into an ordered temporal sequence that reflects structural change across seven decades⁴³. Such trajectories are indicative of demographic evolution because they register how the population pyramid shifts over time, thereby capturing both numerical variation and broader qualitative transformations in demographic conditions¹¹. Figure 2 provides an illustrative example through the trajectories of the United States of America, India, Qatar, and Niger. For the purposes of visualisation, the dimensionality is reduced by Principal Component Analysis to two principal components, so that these demographic paths may be displayed in a two dimensional plane⁴⁴; the methodology section later sets out the procedure by which the dimensionality used for the substantive analysis is reduced to an optimal number of principal components. At this stage, it is sufficient to note that these trajectories are used to represent the manner in which countries move through distinct age structural configurations over time, rather than treating each population pyramid as an isolated cross sectional observation⁴³.

To address the problem from two complementary analytical perspectives, the present study adopts two frameworks: Global Demographic Evolution and Country-Specific Demographic Evolution. Although both frameworks are grounded in the same underlying demographic information, they differ in what is taken as the primary unit of comparison and, consequently, in the type of inference they permit. The Global Demographic Evolution framework is concerned with the changing global configuration of population pyramid structures at successive time points, whereas the Country-Specific Demographic Evolution framework is concerned with the longer-term paths through which those structures change within individual countries^{43,45}. Taken together, they provide both a cross-sectional view of demographic similarity across the globe at a given moment in time, and a longitudinal view of demographic transformation as population structures evolve through time⁴⁵.

In the Global Demographic Evolution framework, the world is examined as a succession of demographic snapshots, each of which provides a representative picture of the global tendency in population pyramid structure at a given moment in time^{2,45}. Hence, at each time instance, countries are grouped according to the similarity of their population pyramid structures at that moment, so that the global clustering pattern may be studied year by year^{31,32}. This makes it possible to ask how many broad demographic groupings are present at different historical points, how their composition changes across time, and how the global demographic landscape evolves as a whole³¹. In this sense, this year-specific clustering procedure is not concerned with the full temporal history of each individual country taken as a single object, but with the changing cross sectional organisation of the world population structure through successive moments⁴⁵. It therefore provides a means of tracing the historical evolution of broad demographic regimes and, by implication, the wider socioeconomic setting reflected in those regimes².

In the Country-Specific Demographic Evolution framework, by contrast, the object of analysis is no longer the single-year population pyramid but the entire trajectory traced by a country through a common analytical space over the period from

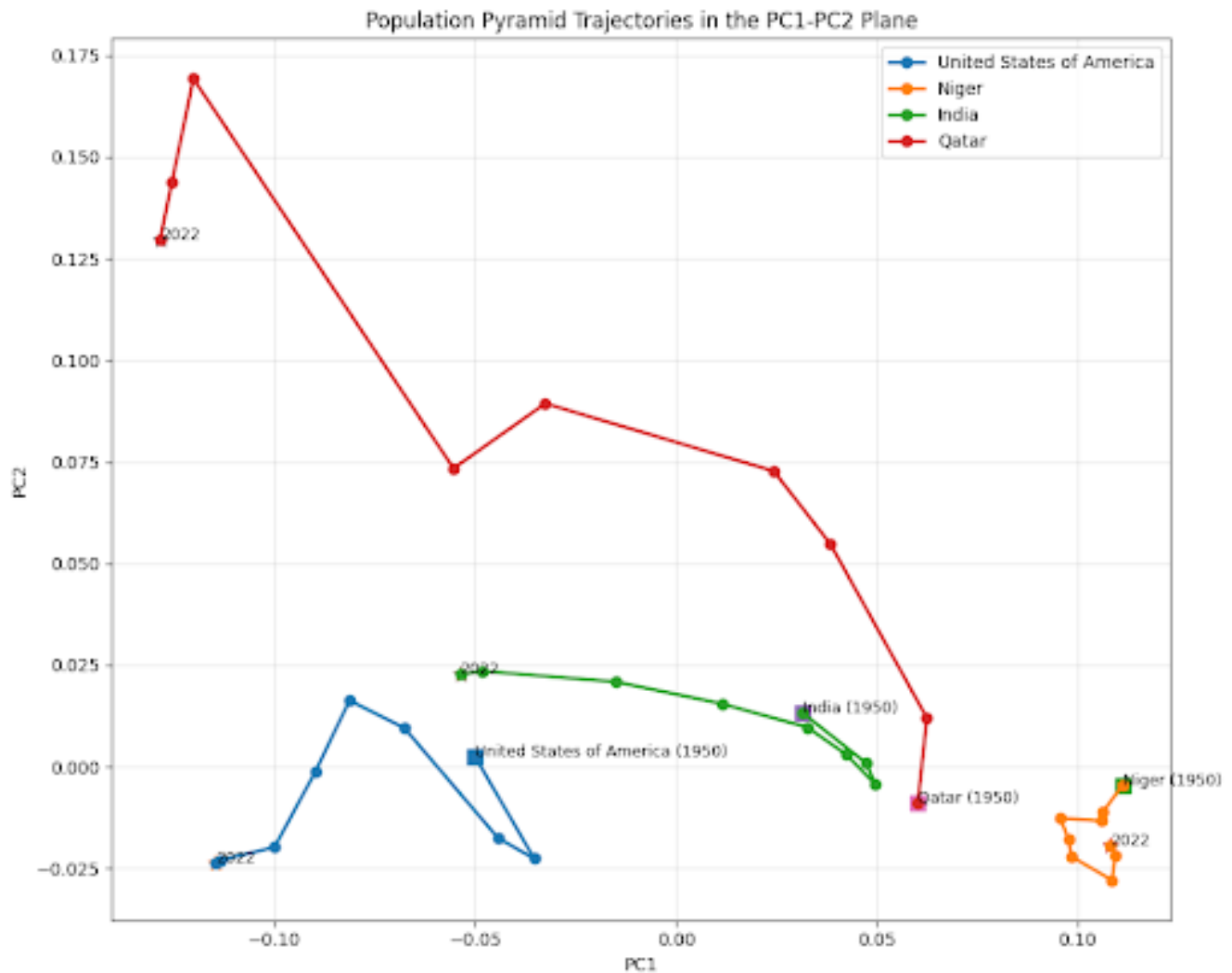


Figure 2. Population pyramid based trajectories of the United States of America, India, Qatar, and Niger in a two dimensional principal component space over the period 1950 to 2020. The figure illustrates how successive population pyramids may be represented as temporal paths within a common analytical frame. For visual clarity, only the first two principal components are shown here, while the full dimensionality reduction procedure used for the substantive analysis is described in the Methodology section.

1950 to 2020^{43,45}. In that setting, the space serves to quantify salient structural qualities of the population pyramid, and each country is represented by the path it traverses through that space across seven decades^{43,44}. More precisely, a single point in this population pyramid trajectory space denotes the nature or structural property of a given country's population pyramid at a particular moment in time, after that pyramid has been reduced to its salient features⁴⁴. These trajectories therefore summarise both quantitative variation and qualitative structural change in demographic conditions, preserving the temporal continuity of population transformation rather than reducing it to isolated observations^{43,45}. Clustering is then performed on the trajectories themselves, so that countries are grouped not because they merely resemble one another at a particular date, but because they exhibit similar patterns in the evolution of their demographic structures^{40,46}. The Country-Specific Demographic Evolution framework thus permits comparison of demographic pathways, and thereby provides a basis for identifying countries whose broader demographic and socioeconomic transitions have unfolded in comparable ways over time⁴⁶.

The methodological novelty of the present study does not lie in the individual use of Principal Component Analysis, spectral clustering, functional smoothing, or Dynamic Time Warping, each of which is already well established^{43,44,47,48}. Rather, it lies in their integration within a unified demographic framework that treats the complete population pyramid as an evolving high-dimensional structure. The framework first represents country-level population pyramids within a common reduced demographic space⁴⁴, then identifies global demographic regimes through repeated cross-sectional clustering^{40,47}, and finally compares the full long-term trajectories of individual countries through temporally aligned, graph-based clustering^{46–48}. This integrated design makes it possible to distinguish between the changing global arrangement of demographic structures and the country-specific pathways through which those structures evolve, thereby providing both a methodological contribution and an empirical account of global demographic transformation.

Accordingly, the study addresses three related questions. First, how has the global configuration of population pyramid regimes changed between 1950 and 2020? Secondly, can countries be grouped according to the form of their long-term demographic trajectories rather than solely by their population structures at individual time points? Thirdly, what additional demographic distinctions become visible when the complete population pyramid is treated as an evolving structure rather than reduced to a small number of conventional summary measures?

Building upon the foregoing discussion, this study makes the following principal contributions:

1. The study treats the population pyramid as a whole demographic structure, rather than reducing it to selected summary indicators.
2. The study presents two complementary frameworks for analysing population pyramids.
 - (a) The Global Demographic Evolution framework examines how the global structure of population pyramids changes through time by clustering countries at successive historical moments.
 - (b) The Country-Specific Demographic Evolution framework examines how each country's population pyramid changes through time by comparing demographic trajectories across seven decades.
3. The study establishes a data-driven framework for identifying both global demographic regimes and country-specific evolutionary pathways from the holistic structure and temporal evolution of population pyramids. This provides a basis for long-term demographic interpretation, planning, and policy development.

Methods

Dimensionality Reduction

Dimensionality reduction was used to obtain a compact representation of the population pyramid data while preserving the principal patterns of variation required for subsequent clustering and visualisation^{44,49}. In this study, Principal Component Analysis was employed to reduce redundancy among the age-group variables and to facilitate the analysis of demographic structure in a lower-dimensional space⁴⁴.

Principal Component Analysis (PCA)

Principal Component Analysis was used to obtain a compact representation of the population pyramids while retaining the dominant variation in age structure⁴⁴. Each country-year observation was represented by the normalised proportions of the population in 21 five-year age groups. Because neighbouring age cohorts are strongly correlated, direct analysis in the original feature space contains considerable redundancy. PCA addresses this by transforming the age-group variables into a smaller set of uncorrelated components ordered according to the proportion of variance they explain.

Let $X \in \mathbb{R}^{n \times p}$ denote the centred data matrix, where each row corresponds to a country-year observation and each column corresponds to one of the ($p=21$) age-group proportions. The reduced representation is given by,

$$Z = XW_k \quad (1)$$

where W_k contains the first k principal components and Z contains the corresponding component scores. The number of retained components was selected as the smallest value of k for which the cumulative explained variance reached at least 95 per cent. In the present analysis, this criterion retained four principal components.

The first principal component was examined separately because its loading structure provided an interpretable summary of demographic maturity, contrasting younger and older age groups. The full set of retained components was then used to construct multivariate country trajectories for the Country-Specific Demographic Evolution analysis. Thus, PCA served both as an interpretable representation of the dominant ageing-related pattern and as a compact multivariate basis for clustering the longitudinal evolution of complete population pyramids.

Functional Data Analysis

Functional Data Analysis was employed to represent each country's demographic evolution as a continuous temporal process rather than as a sequence of isolated annual observations⁴³. In the present study, this formulation was applied to the first principal component, which summarises the dominant contrast between youthful and mature population structures. Accordingly, the annual PC1 scores recorded for each country from 1950 to 2020 were treated as observations of an underlying smooth demographic trajectory.

For country c , the observed PC1 score at year t may be written as,

$$y_{c,t} = g_c(t) + \varepsilon_{c,t} \quad (2)$$

where $g_c(t)$ denotes the underlying demographic trajectory and $\varepsilon_{c,t}$ represents short-term variation or residual error. Estimating $g_c(t)$ makes it possible to examine the direction, pace, and curvature of demographic change while reducing the influence of minor year-to-year irregularities. The resulting functions were subsequently used in the Country-Specific Demographic Evolution analysis, where countries were compared according to the overall form of their long-term demographic trajectories.

Cubic Spline Approximation

A cubic smoothing spline was fitted separately to the PC1 series of each country^{50,51}. The fitted trajectory consists of piecewise cubic polynomials joined at a set of knots. Within interval $[t_i, t_{i+1}]$ the spline may be expressed as,

$$S_i(x) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3 \quad (3)$$

where a_i , b_i , c_i , and d_i are the coefficients associated with the i^{th} interval. Adjacent polynomial segments are constrained to have equal function values, first derivatives, and second derivatives at their common knots, thereby producing a smooth and twice-differentiable trajectory. The spline estimate balances agreement with the observed PC1 scores against excessive temporal variation. This may be represented by the penalised objective,

$$\min_{g_c} \sum_{t=1}^T (y_{c,t} - g_c(t))^2 + \lambda \int (g_c''(u))^2 du \quad (4)$$

where the first term measures the discrepancy between the observed and fitted values, while the second penalises excessive curvature. The parameter λ governs the trade-off between fidelity and smoothness.

In the implementation, cubic splines of degree three were fitted using a smoothing factor of $s = 1$. Each fitted function was evaluated at 71 annual points covering the period from 1950 to 2020. This procedure preserved the principal long-term form of demographic transition while reducing the influence of short-term fluctuations, thereby providing a continuous functional representation suitable for subsequent trajectory clustering.

Clustering Algorithms

Clustering was used to identify groups of countries with similar population pyramid structures or comparable long-term demographic trajectories without imposing predefined demographic categories. The study employed spectral clustering because the resulting groups were not assumed to be linearly separable in the original feature space, while k-means was used only to obtain the final discrete partition within the spectral embedding.

Spectral Clustering

Spectral clustering was used to identify groups from the similarity structure of the demographic observations^{47,52}. Each country was represented as a node in a weighted graph, and the edge weight between countries i and j reflected the similarity of their population pyramid representations or demographic trajectories.

Given a pairwise distance d_{ij} the simimatrix W was constructed using a Gaussian kernel:

$$W = \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right) \quad (5)$$

where σ controls the rate at which similarity decreases with distance. For the Global Demographic Evolution analysis, d_{ij} was calculated from the reduced population pyramid representations at a given year. For the Country-Specific Demographic Evolution analysis, it was obtained from the distance between longitudinal demographic trajectories.

The degree matrix D was defined as

$$D_{ii} = \sum_j w_{ij} \quad (6)$$

and the graph Laplacian was then constructed as

$$L = D - W \quad (7)$$

The eigenvectors associated with the smallest eigenvalues of L provide a lower-dimensional representation of the graph in which strongly connected observations are placed near one another. The final country groups were obtained by applying k-means to this spectral embedding.

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$$\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n \quad (8)$$

The preferred number of clusters (k) was identified from the largest relevant difference between successive eigenvalues,

$$k = \arg \max_i (\lambda_{i+1} - \lambda_i) \quad (9)$$

This procedure allowed the principal cluster structure to be inferred from the connectivity of the demographic similarity graph rather than fixed in advance.

Standard K-Means Clustering

K-means clustering was applied to the spectral embedding to obtain the final discrete partition of countries^{53,54}. The number of clusters K was determined beforehand using the eigengap criterion, after which the embedded observations were assigned to groups by minimising the within-cluster sum of squared distances:

$$\min_{\{c_j\}_{j=1}^k} \sum_{j=1}^k \sum_{\mathbf{z}_i \in C_j} \|\mathbf{z}_i - \mu_j\| \quad (10)$$

where $\mathbf{z}_i$ denotes the spectral representation of observation i , C_j denotes the j^{th} cluster, and μ is its centroid. In the present study, k-means therefore served only as the final assignment stage of the spectral clustering procedure and was not applied independently to the original demographic variables.

Data Collection

The population data required for the analysis were obtained from the United Nations Department of Economic and Social Affairs, Population Division's 2024 Revision of the World Population Prospects⁴². The extracted variables comprised the geographical entity (country or area), the year (1950-2020), five-year age cohorts ranging from "0-4" to "100+", and absolute population counts stratified by biological sex (Male and Female).

Data Preprocessing

The data preprocessing phase was conducted to refine the dataset for a country-level longitudinal analysis. The geographical entity variable was cleaned by systematically removing regional, continental, and developmental aggregates. The resulting geographical entity variable comprised 201 unique locations, including recognized sovereign states, special administrative regions (e.g., China, Hong Kong SAR), non-sovereign territories (e.g., Puerto Rico and Guam), and overseas departments (e.g., Réunion), as defined by the United Nations Population Division. For readability, the term country is used hereafter to refer to all geographical entities included in the analysis. A total population variable was derived by aggregating male and female population counts and normalising them to percentages by dividing each age count by the aggregate country-year total. A total population variable was derived by aggregating male and female population counts and normalizing them to percentages by dividing each age count by the aggregate country-year total.

Global Demographic Evolution

The Global Demographic Evolution framework was introduced to examine the demographic structure of the world as a changing whole, rather than focusing only on individual country trajectories. In this approach, the population pyramid is used as the principal instrument for grouping countries according to their age-structural similarity at each historical moment. The purpose is therefore to observe how these groups appear, separate, persist, contract, expand, or reorganise through time, thereby providing a cross-sectional record of how countries are configured around one another within the global demographic landscape.

The value of this perspective lies in its ability to trace broad historical transformation. A population pyramid is not merely a demographic summary, but also an indirect expression of the economic, social, epidemiological, and institutional conditions through which a population has passed. When countries are clustered at successive time points on the basis of pyramid structure, the resulting sequence of partitions reveals how major demographic regimes have been reconstituted from the mid twentieth century to the contemporary period. In this way, the method views the globe as a whole through time, mapping global demographic evolution through the formation, separation, and redistribution of country clusters.

Methodologically, each year is treated as a demographic snapshot of the world. Countries are clustered according to the similarity of their population pyramid structures in that year, and the resulting yearly partitions are then read in sequence. This makes it possible to examine not only which countries resemble one another at a given moment, but also how the global organisation of demographic forms changes across successive moments. The framework therefore transforms a set of cross-sectional observations into an interpretable record of global demographic reordering.

Spectral clustering was used for this purpose because population pyramid similarities need not form simple linear groups in the reduced feature space. Countries may instead be related through broader patterns of connectivity, where groups are defined by the strength of their structural resemblance within the similarity graph. Spectral clustering is well suited to recovering such organisation, allowing the cluster structure to arise from the internal configuration of the data rather than from a prespecified demographic classification.

Nevertheless, this perspective remains only one side of the analytical design. It reveals how the global pattern of demographic groupings changes from one time point to another, but it does not by itself preserve the continuous pathway through which a particular country arrives at its demographic position.

Country-Specific Demographic Evolution

Whereas the Global Demographic Evolution framework examines the world as a sequence of cross sectional demographic configurations, the Country-Specific Demographic Evolution framework shifts attention from the global snapshot to the country specific path. The Global Demographic Evolution approach is informative for identifying how the composition of demographic groupings changes from one historical moment to another, yet it does not by itself preserve the continuous route through which a country arrives at its present structural position. For that reason, the analysis is extended here to trajectory clustering, in which the object of comparison is no longer the isolated population pyramid observed at a single year, but the ordered evolution of population pyramid structure through time. In this way, the methodology moves from the study of changing global arrangements to the study of long run demographic pathways.

As noted earlier in the introduction, a population pyramid based trajectory may be understood as the path traced by a country's age structure through a common analytical frame over the period from 1950 to 2020. Dynamic clustering is founded precisely upon this idea. Its purpose is to compare countries not merely in terms of their demographic resemblance at particular dates, but in terms of the manner, pace, and direction in which their population structures have evolved across seven decades. This perspective is methodologically important because demographic transition is inherently a process unfolding through time. Two countries may display similar pyramid shapes in one year and yet have reached that apparent similarity through markedly different historical sequences. A trajectory based framework therefore permits comparison of demographic pathways rather than demographic states alone, and thereby offers a more faithful account of long term structural change.

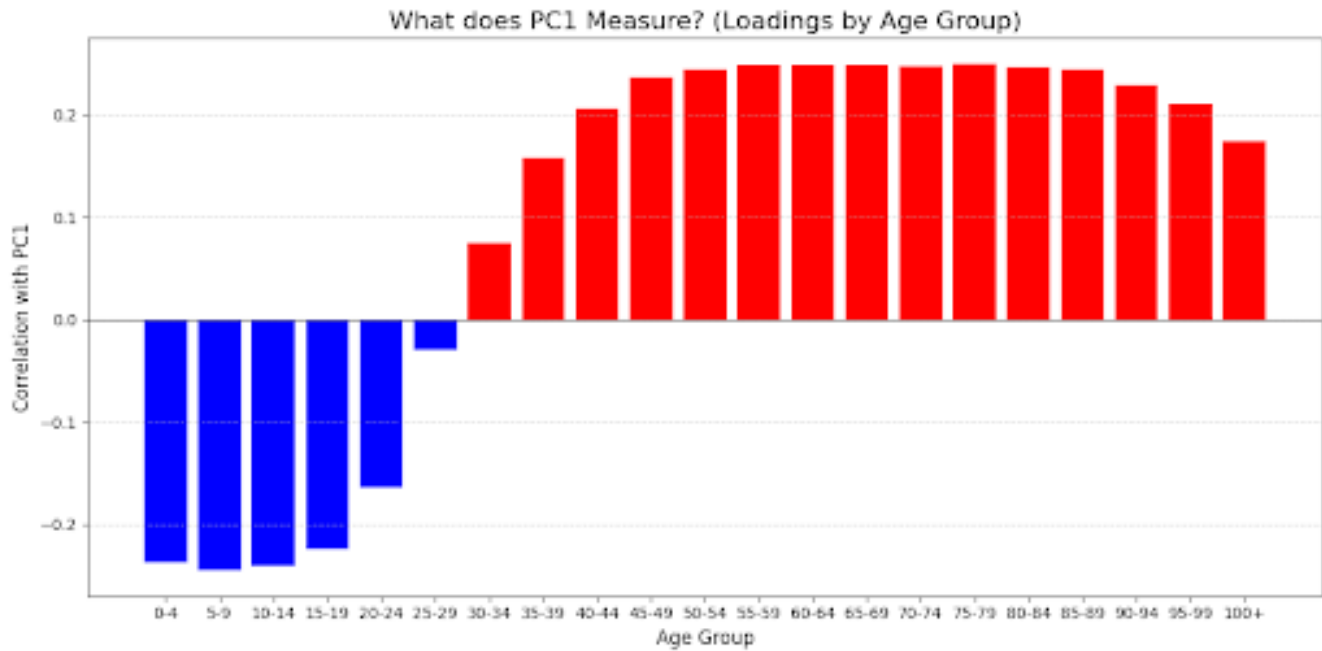


Figure 3. Loadings of the first principal component (PC1) across 5-year age cohorts.

Two Representations of Country-Specific Demographic Evolution

For this purpose, two complementary dynamic formulations were adopted. The first, here termed Univariate Dynamic Clustering, deliberately reduces demographic evolution to the first principal component, which functions as an age structure transition index and allows each country's long run movement to be interpreted along a single dominant axis of demographic change. The second, here termed Multivariate Dynamic Clustering, retains the optimal number of principal components required to preserve the great majority of structural variation in the population pyramids, thereby representing each country through a richer multivariate trajectory. The justification for employing both formulations is substantive as well as methodological. The univariate approach offers parsimony and interpretability, making it possible to examine the broad direction and velocity of demographic transition in an especially transparent form. The multivariate approach, by contrast, guards against excessive reduction by retaining secondary but demographically meaningful features of population pyramid shape that are not fully captured by a single component. Taken together, the two approaches allow the analysis to proceed from an interpretable one dimensional summary of demographic evolution to a fuller representation of the underlying structural transformation.

PC1 as a Demographic Maturity Index

The dataset, comprising 201 geographical entities over the period from 1950 to 2020, was structured as a matrix, with each row representing a unique country-year observation. The population was distributed across 21 chronological age cohorts (ranging from 0–4 to 100+) for each observation. Before PCA was applied to reduce them to a single informative dimension, these cohorts were normalized to percentages to capture the relative age structure (proportions) rather than absolute population size. The 21 percentage-based variables were standardized column-wise to a mean of zero and unit variance, ensuring equal contribution of each age cohort to the variance.

The first principal component (PC1) can be recognised as a Demographic Maturity Index (DMI) as justified by the bipolar loading structure (Figure 3). PC1 quantifies age disparity, capturing the see-saw effect inherent in population structure as the percentages of all age cohorts sum to 100%. The proportional expansion of one life stage is inversely tied to the shrinking of another. PC1 leverages a pivot point at age 30, where cohorts aged 0 to 29 exhibit negative loadings, while those aged 30 to 100+ exhibit positive loadings. Because the 21 age variables were standardized to Z-scores, a high positive PC1 score indicates a profile in which the proportions of cohorts aged 30 to 100+ are significantly above the global historical mean, while those under 30 are significantly below it. That is, a country achieves a high positive PC1 value when it must possess a youth deficit ($Z < 0$ for cohorts under 30) and an elderly surplus ($Z > 0$ for cohorts 30 years and older).

Functional Data Transformation via Cubic Splines

The discrete PC1 scores for each country [Figure 4] were transformed into a continuous temporal process using univariate B-splines. Cubic splines ($k = 3$) were employed for the longitudinal data (1950–2020) to ensure high smoothness and twice-

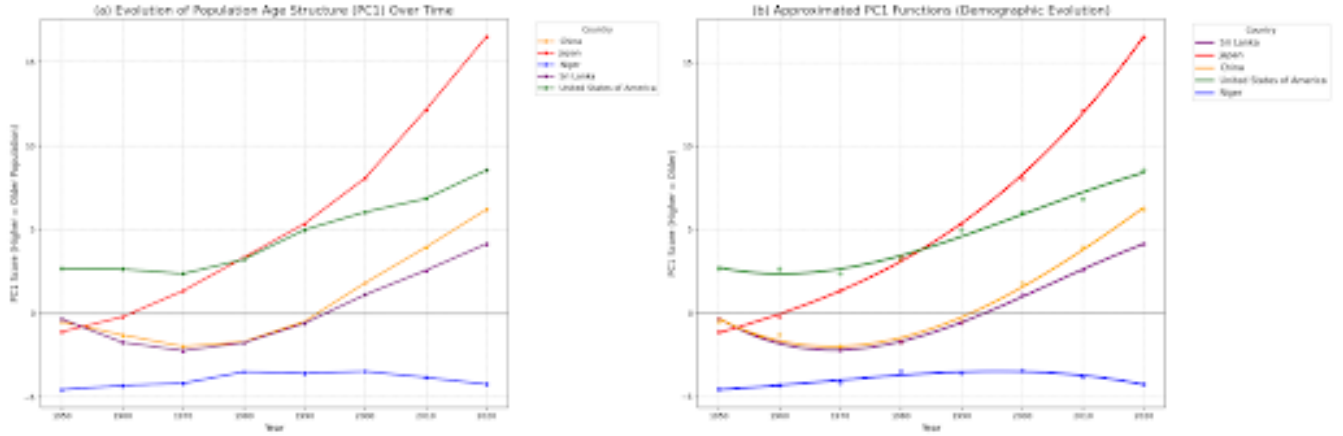


Figure 4. Evolution and Functional Approximation of the first Principal Component (PC1).

differentiability, which are essential for capturing the underlying velocity of demographic trends. The approximation was performed over a dense evaluation grid of 71 points. A smoothing parameter of $s=1$ was used to filter out high-frequency noise and short-term demographic fluctuations, while strictly preserving the long-term nonlinear trends inherent in population aging. The resulting functions [Figure 4] helped interpret the PC1 scores not only as a sequence of points but also as a dynamic aging trajectory unique to each country.

Spectral Clustering followed by Standard K-Means Clustering

The 201 geographical entities, based on their demographic evolution from 1950 to 2020, were clustered using spectral clustering on a dense evaluation grid of 71 points sampled from the cubic spline approximations of PC1. Cubic splines serve as a preprocessing step, filtering out high-frequency noise and short-term demographic shocks that might otherwise distort direct PC1 measurements. Spectral clustering was selected for its ability to identify complex, nonlinear similarities in demographic trajectories by projecting the functional data onto a lower-dimensional subspace derived from the graph Laplacian.

An eigengap heuristic analysis was performed across a range of Gaussian kernel scales (sigma). This approach identified the most stable cluster counts (k) by maximizing the gap between consecutive eigenvalues of the Laplacian matrix. Then, the standard K-means algorithm was utilized to perform the final clustering in the transformed space.

Multiple-PC Representation of Country-Specific Demographic Evolution

Whereas the preceding univariate formulation deliberately isolates the dominant mode of demographic variation, the present formulation recognises that the evolution of a population pyramid cannot always be represented adequately by PC1 alone. Although PC1 provides an interpretable summary of the broad contrast between younger and older age structures, it may obscure secondary but demographically meaningful features, including the relative balance of intermediate age groups, the concentration of working-age cohorts, and finer variations in the morphology of transition. For this reason, the analysis was extended to a multivariate trajectory representation in which each country was described by the leading principal components rather than by a single ageing-related index.

The retained dimensionality was selected using the cumulative explained variance criterion described earlier. This procedure preserves the principal modes through which population pyramids differ across countries and across time, while avoiding the redundancy and noise associated with the full original age-cohort space. In the present study, the 95 per cent explained variance threshold yielded four retained principal components. Thus, each country at each time point was represented by a four-dimensional principal component vector, and the ordered sequence of these vectors defined that country's demographic trajectory:

$$\tau_c = [z_{c,1}, z_{c,2}, \dots, z_{c,T_c}] \quad (11)$$

where τ_c denotes the demographic trajectory of country c , $z_{c,t}$ denotes the reduced dimensional principal component representation of that country at time t , and T_c denotes the total number of temporal observations available for country c . Thus, τ_c represents the ordered sequence of low dimensional demographic states through which country c passes over the study period.

For visual exposition, trajectories are displayed in the plane defined by the first two principal components, since this provides an intelligible graphical summary of movement through the reduced demographic space (Figure 2). However, the

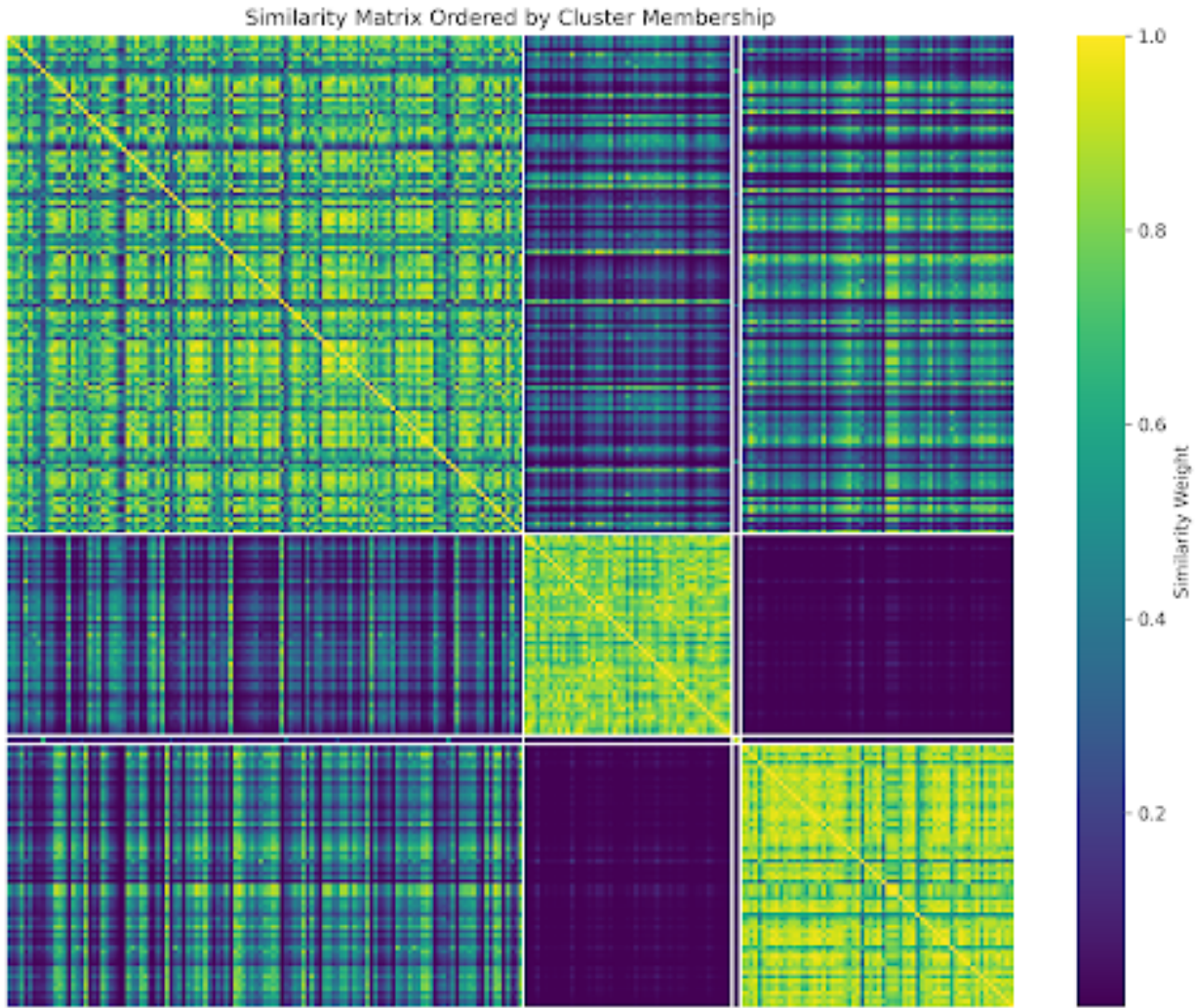


Figure 5. Heatmap representation of the similarity matrix constructed from Dynamic Time Warping distances between country specific demographic trajectories. Each element w_{ij} is encoded by colour intensity according to the Gaussian kernel transformation defined in Equation (5). The matrix is ordered by final cluster membership to facilitate visual inspection of the underlying similarity structure.

clustering analysis itself was carried out using the full retained four-dimensional representation rather than the two-dimensional visual projection.

Once the multivariate trajectories had been constructed, pairwise dissimilarities between countries were computed using Dynamic Time Warping^{46,48}. This was necessary because countries may undergo similar demographic transitions at different historical moments or at different rates. Dynamic Time Warping permits flexible temporal alignment, allowing countries to be compared according to the overall form of their demographic pathways rather than by exact synchrony in calendar time.

The resulting Dynamic Time Warping distances were transformed into an affinity matrix using a Gaussian kernel as given in equation (5). Here d_{ij} denotes the Dynamic Time Warping disparity between countries i and j . This affinity matrix represents the graph structure on which spectral clustering was performed. Countries with similar long-term demographic trajectories receive stronger connections in this graph, while countries with dissimilar trajectories receive weaker connections.

The number of clusters was determined from the eigenspectrum of the graph Laplacian rather than fixed in advance. The eigenvalue and eigengap plots (Figure 6) indicated a four-cluster solution, suggesting that the similarity graph contains four principal connectivity structures. Spectral clustering was then applied to the affinity matrix using this cluster number. In this way, the final grouping was based on the internal organisation of multivariate demographic trajectories, rather than on predefined demographic categories or isolated cross-sectional resemblance.

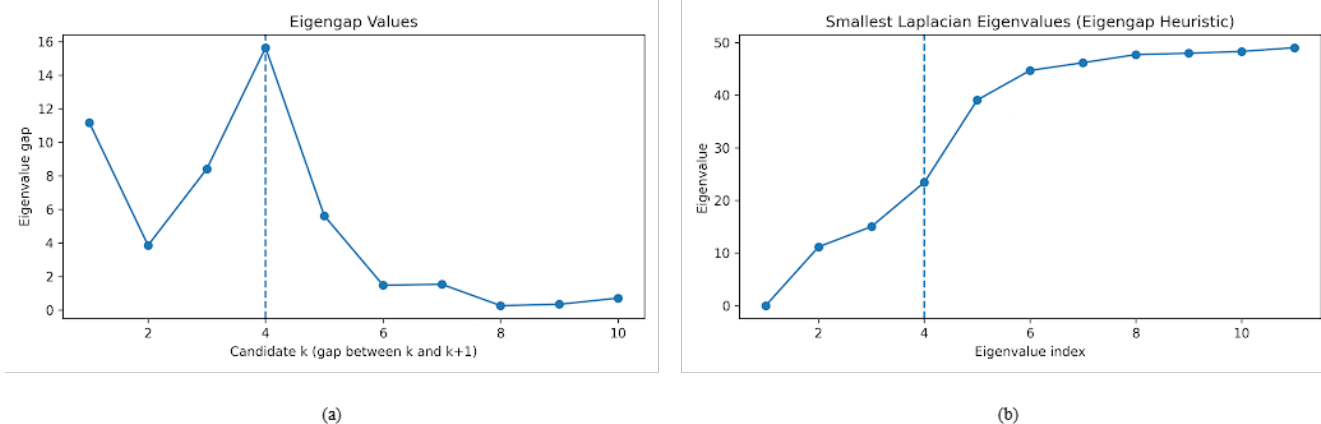


Figure 6. Graph Laplacian spectrum and eigengap analysis for multivariate population pyramid trajectories. (a) Ordered eigenvalues of the graph Laplacian constructed from the trajectory affinity matrix, summarising the connectivity structure of the country similarity graph. (b) Eigengap profile, showing successive differences between adjacent eigenvalues, with the largest relevant gap indicating the preferred number of trajectory clusters.

Results

Global Demographic Evolution

The Global Demographic Evolution analysis was conducted by clustering country-level population pyramids separately at each decadal time point from 1950 to 2020. For each year, the number of clusters was determined from the eigengap criterion, and the resulting country groupings were subsequently labelled according to their aggregate age-structural characteristics. The assigned labels therefore provide descriptive summaries of the demographic regimes identified within each year, rather than predefined classes imposed prior to clustering.

Figure 7 presents the temporal variation in the optimal number of global demographic clusters. The number of clusters varied across the study period, ranging from three to seven. Five clusters were identified in 1950, followed by seven clusters in 1960, six clusters in 1970, and seven clusters in 1980. The smallest number of clusters was observed in 1990, where the eigengap criterion supported a three-cluster solution. Thereafter, the number of clusters increased to seven in 2000, before declining to six in 2010 and five in 2020. Thus, the results indicate that the cross-sectional structure of global demographic regimes did not remain constant over time, but changed in both the number and composition of identified groups.

Figure 8 shows the corresponding spatial distribution of the labelled demographic regimes for each decade. Each panel represents the country groupings obtained for a single year, with the regime labels derived from the average age composition of the countries assigned to each cluster. Across the panels, the maps show the decadal configuration of global population pyramid regimes, including broad-base youthful, transitional, mature ageing, and, in some years, working-age concentrated regimes. The maps also show that the number of regimes differs by year, in accordance with the cluster counts reported in Figure 7.

Taken together, Figure 7 and Figure 8 report the cross-sectional evolution of global demographic structure from 1950 to 2020. Figure 7 summarises the changing number of demographic regimes supported at each decade, while Figure 8 displays how the corresponding regimes are distributed geographically. These results provide the empirical basis for examining how countries are configured around one another through population pyramid similarity at successive historical moments.

Country-Specific Demographic Evolution

PC1-Based Country-Specific Demographic Evolution

To determine the optimal number of clusters for the demographic analysis, the eigengap heuristic was evaluated across a range of Gaussian kernel scales (σ). The selection of optimal cluster numbers ($k = 2$, $k = 3$, and $k = 5$) was based on mathematical stability, as assessed by the persistence of eigengap peaks and structural resolution. $k=2$, characterized by a large eigengap across a wide range of σ values, is the most stable partition. $k = 3$ provides a balanced intermediate resolution with consistent stability in a wide range of σ values, while $k = 5$ offers the highest stable resolution before the eigengaps become increasingly volatile. This hierarchical approach allows exploration of demographic history at three distinct levels: a binary macrostructure, a tripartite intermediate transition, and a granular five-stage classification of aging motifs.

The average functional motifs for each cluster were derived by identifying the centroid archetype that characterizes the group's collective movement over time. The average motif value for each specific year t was determined by calculating the arithmetic mean of the scores for all N countries within that cluster for that year. These motifs were plotted (Figure 9) to

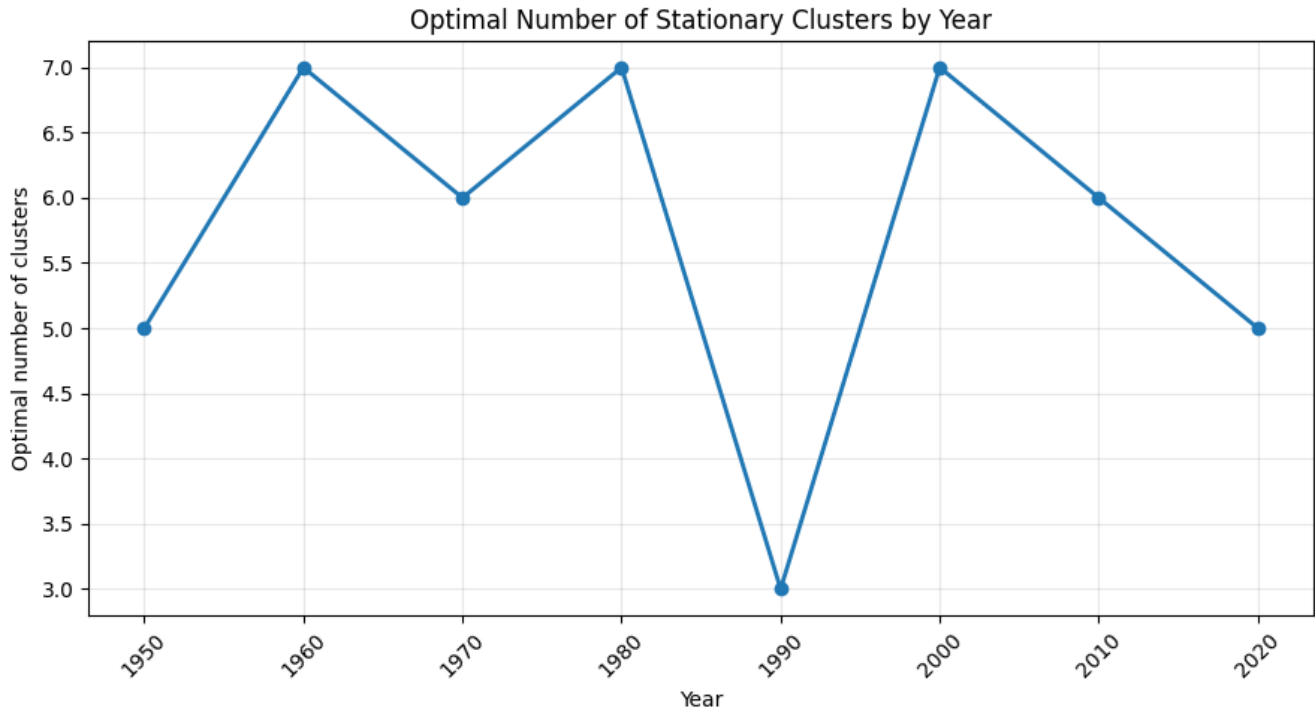


Figure 7. Evolution of the optimal number of global demographic clusters from 1950 to 2020. The number of clusters at each decadal time point was selected using the eigengap criterion applied to the yearly population pyramid similarity structure.

transform the abstract mathematical partitions of $k = 2$, $k = 3$, and $k = 5$ into interpretable demographic stages, visually assessing the distinct aging archetypes based on their unique starting positions, velocities, and acceleration profiles.

At the macro-level of resolution ($k = 2$), the dataset is partitioned into two primary archetypes. The older population motif comprises 107 geographical entities. The younger population motif comprises 94 entities. The older population motif trajectory originates at the global historical mean in 1950 and accelerates in the maturing direction over the 71 years. This motif is predominated by entities from Europe, Australasia, North America, and East Asia, including Japan, Italy, Germany, and the United States. The younger population motif maintains a young surplus persistently. It begins at a significantly lower maturity level and follows a shallow-basin path, descending further into proportional youthfulness, followed by a maturing phase, still ending with a young surplus by 2020. This motif is predominated by entities from Sub-Saharan Africa, South Asia, and the Middle East, such as Niger, Nigeria, Afghanistan, and Pakistan. This macro-scale classification distinguishes a distinct bifurcation between the world's two most populous entities, with China assigned to the older population motif and India assigned to the younger population motif.

At the tripartite level of resolution ($k = 3$), the dataset is partitioned into three archetypes: the young population motif (72 entities), the transitioning population motif (66 entities), and the old population motif (63 entities). The old population motif trajectory starts above the $PC1 = 0$ in 1950 and ascends, reaching the peak value by 2020. This cluster predominates entities from Europe, North America, Japan, and Australasia, including the United Kingdom, Germany, and New Zealand. The transitioning population motif trajectory starts at a youthful maturity level and follows a high-velocity upward path. It is the only cluster at this resolution to cross the zero-line pivot point. Regionally, this motif is composed of emerging economies across East Asia, Latin America, and the Middle East, such as China, India, Brazil, Mexico, and Turkey. Conversely, the young population motif trajectory begins at the extreme youthful maturity level, follows a U-shaped pattern, descending to its lowest maturity level in around 1995 before a modest recovery by 2020. This group is distinguishably represented by nations in Sub-Saharan Africa and parts of Central and South Asia, including Niger, Nigeria, Ethiopia, and Pakistan. At this resolution, the world's two most populous nations, China and India, both converge within the transitioning population motif.

At the highest stable resolution ($k = 5$), the dataset is partitioned into five archetypes that characterize the relatively micro-level global demographic evolution. The persistent youth surplus motif (55 entities) maintains the most youthful structure, originating at a low maturity level and following a shallow-basin path that remains deeply negative through 2020. This group predominates Sub-Saharan Africa and a few parts of the Middle East, including Niger, Nigeria, Ethiopia, and Iraq. The U-rebound towards aging motif (35 entities) has a non-monotonic trajectory. It starts near zero-line in 1950, experiences a

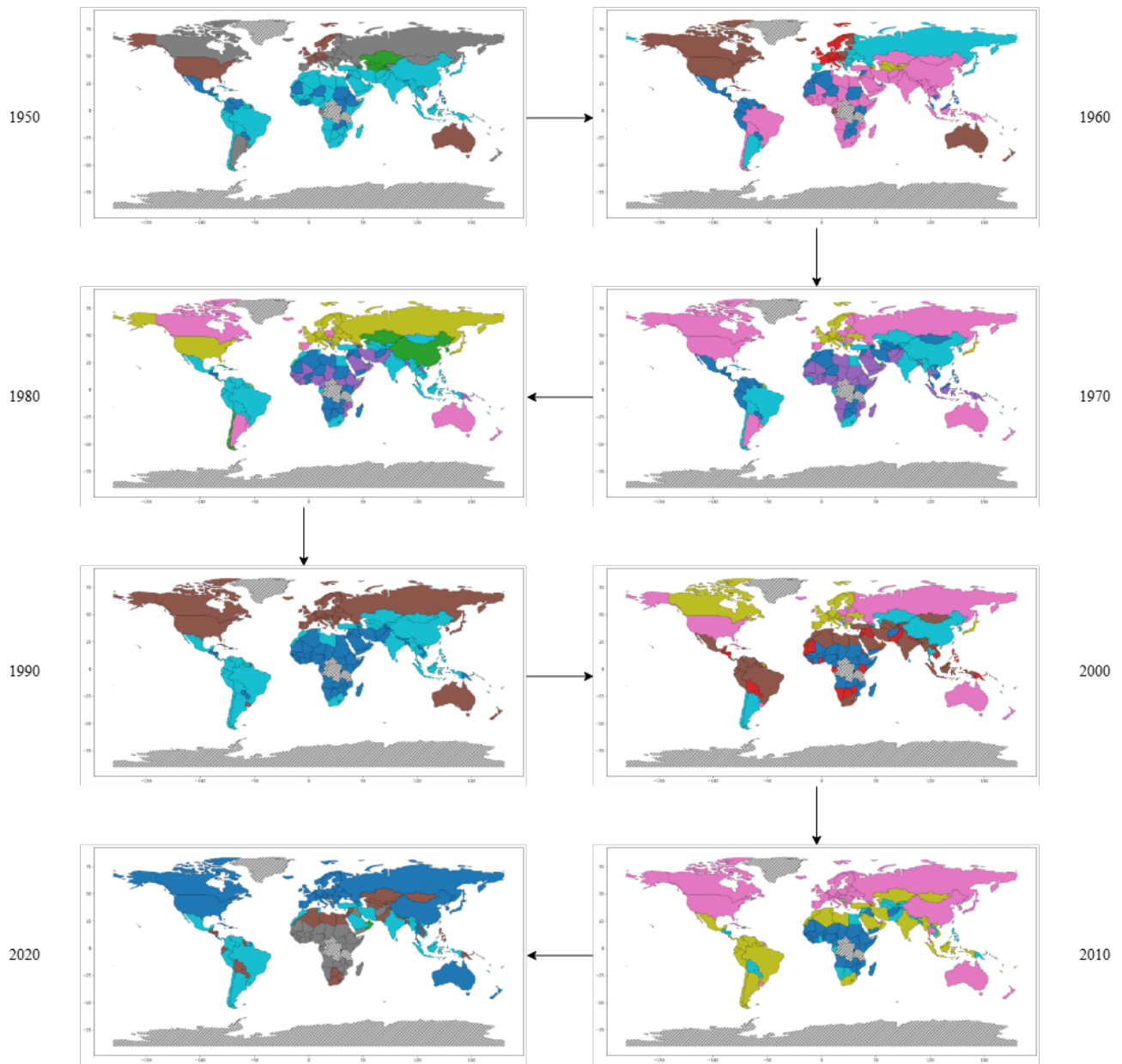


Figure 8. Spatial distribution of global demographic regimes from 1950 to 2020. Each panel shows the labelled population pyramid regimes obtained by clustering countries separately at each decadal time point. Regime labels were assigned after clustering according to the average age-structural composition of each cluster and should therefore be interpreted within each year.

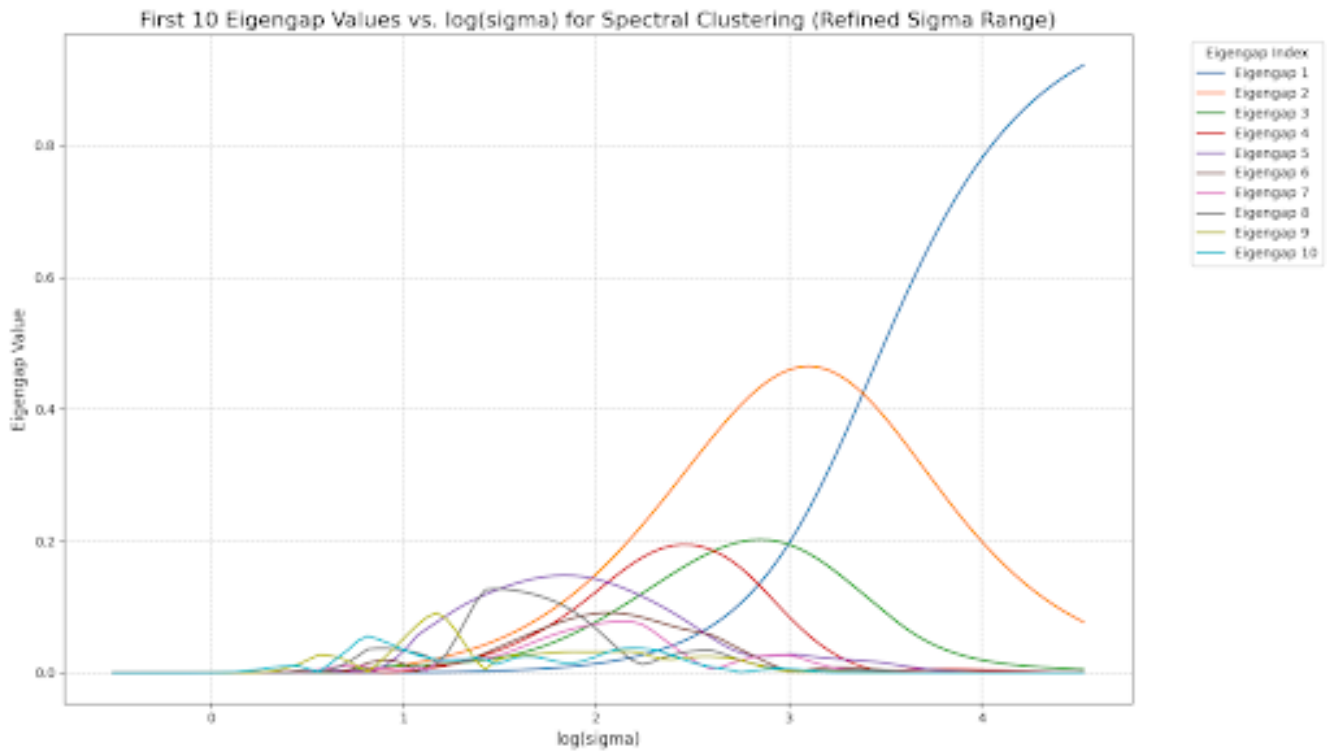


Figure 9. Eigengap Heuristic Analysis for Optimal Cluster Selection Across Varying Gaussian Kernel Scales.

proportional younging descent to a trough by 1985, and subsequently demonstrates high positive acceleration towards maturity, crossing the zero-line by 2015. This cluster comprises nations from Central and South Asia and Africa, such as India, Pakistan, Kazakhstan, and South Africa.

The young surplus to aging: J curve ascent motif (47 entities) represents the most aggressive structural transformation. It begins extremely youthful and breaking away after 1970 to crossing the zero-line in approximately 2005, ending the period in a high maturity level. This motif is dominated by emerging economies in East Asia and Latin America, including mainland China, Brazil, Mexico, and Turkey. The post-transitional aging ascent motif (38 entities) originates near the zero-line and follows a steady, near-linear climb to a high positive PC1. Regionally, this archetype includes North America and Australasia, featuring entities such as the United States, Canada, Australia, and New Zealand. The persistent aging ascent motif (26 entities) represents the most matured populations throughout 1950-2020, with an exponential upward-facing parabola. This motif is exclusively composed of Japan and European nations such as Germany, Italy, and the United Kingdom.

When increasing resolution from the binary macro-structure ($k = 2$) to the tripartite level ($k = 3$), the younger population motif ($n = 94$) is refined as 72 entities remained in the young population motif, while 22 entities with compressed transition, including India, Pakistan, and Egypt, shifted into the new transitioning population motif. The macro-level older population motif ($n = 107$) is partitioned, with 63 entities retained in the old population motif and 44 entities that reached maturity more recently, such as mainland China, Brazil, and Mexico, reallocating to the transitioning motif ($n=66$).

When zooming in from $k = 3$ to $k = 5$, within the young population motif ($n = 72$), 55 entities remain as the persistent youth surplus, including Niger, Afghanistan, and Ethiopia, while 17 entities, such as Gabon, Myanmar, and Nepal, shift into the U-rebound towards aging motif. The transitioning population motif ($n=66$) undergoes a three-way partition as 47 entities, including mainland China, Brazil, Mexico, and Turkey, shifting to young surplus to aging: J-curve ascent motif; 18 entities, such as India, Pakistan, and South Africa, join the U-rebound towards aging motif; and a single entity, Chile, places in the post-transitional aging ascent motif. The old population motif ($n = 63$) is bifurcated: 37 entities, including the United States, Australia, and the Russian Federation, are reallocated to the post-transitional aging ascent motif, while 26 entities, such as Japan, Germany, and Italy, form the persistent aging ascent motif.

Multiple-PC Country-Specific Demographic Evolution

Principal Component Representation of Population Pyramids Application of Principal Component Analysis to the normalised population pyramids yielded a compact representation of demographic structure while preserving the dominant

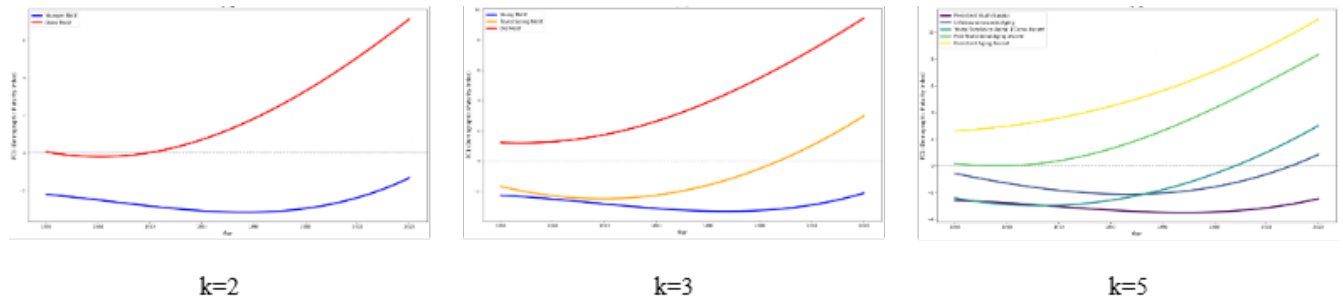


Figure 10. Average Functional Motifs of Clusters at 3 Hierarchical Resolutions.

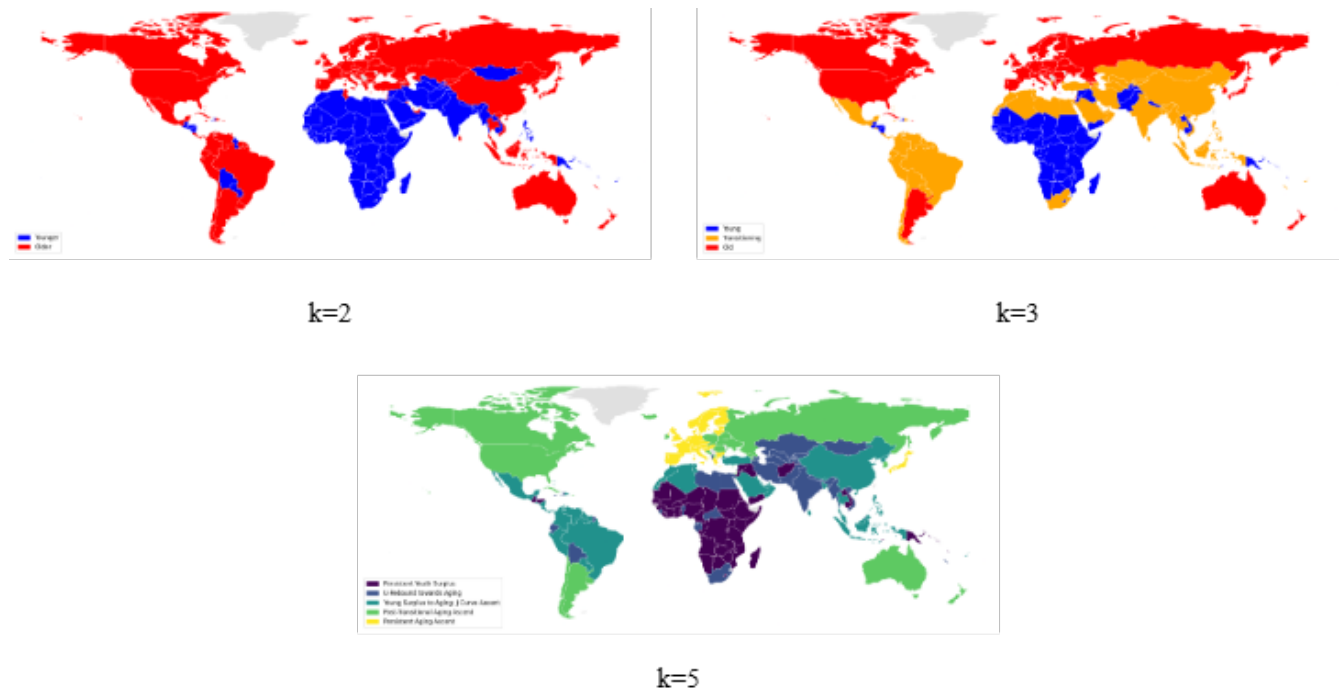


Figure 11. World maps to distinguish the country distribution of clusters at 3 Hierarchical Resolutions.

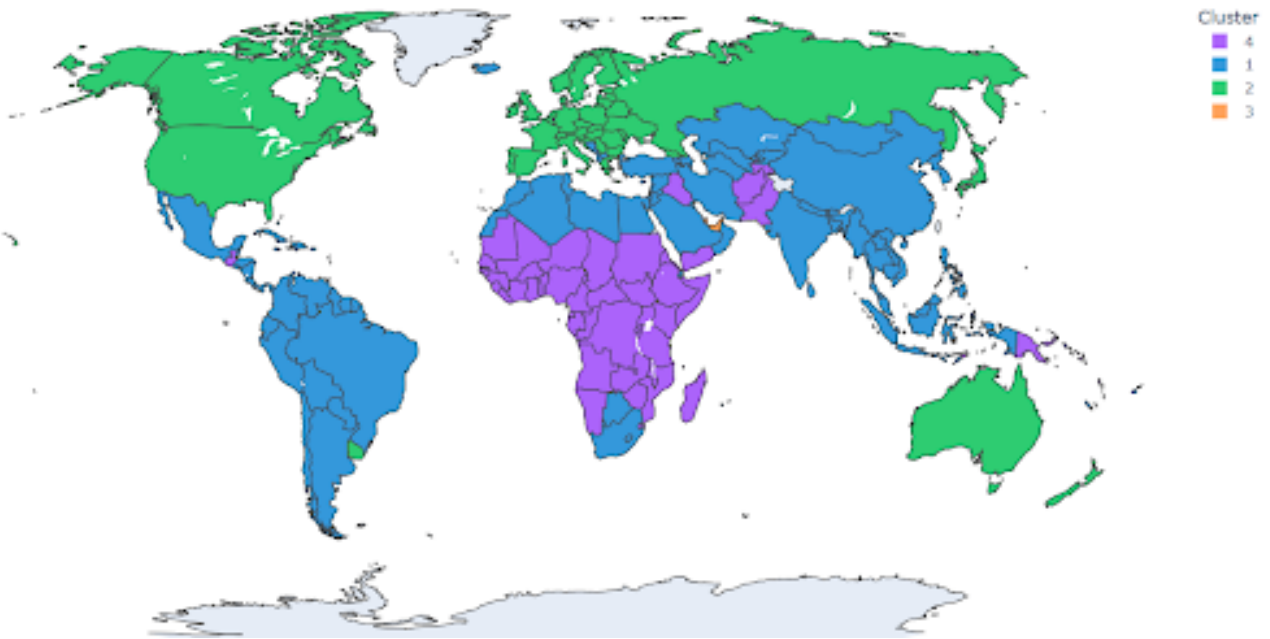


Figure 12. Enter Caption

patterns of variation in the data. The dimensionality reduction procedure retained the smallest number of principal components whose cumulative explained variance exceeded ninety five per cent of the total variance. Each country year observation was therefore represented by a vector of principal component scores summarising the age structure of the population.

This transformation enables meaningful comparison of demographic evolution across countries by capturing the essential shape characteristics of the population pyramid while eliminating redundancy across age groups.

Similarity Structure of Demographic Trajectories Each country was represented by a chronological sequence of principal component vectors, thereby forming a multivariate demographic trajectory describing the temporal evolution of its population structure. Pairwise similarities between these trajectories were computed using Dynamic Time Warping distances followed by a Gaussian kernel transformation to obtain a weighted similarity matrix.

The structure of this similarity matrix is illustrated in the heatmap representation shown in Figure 5. When ordered according to cluster membership, the matrix exhibits a pronounced block diagonal configuration. Regions along the diagonal display relatively high similarity values, indicating strong structural resemblance among countries belonging to the same group. In contrast, the off diagonal regions are characterised by substantially lower similarity values, suggesting weaker connections between different groups of countries.

The emergence of such block structure in the similarity matrix indicates that the demographic trajectories of countries do not form a homogeneous continuum, but instead exhibit natural groupings in the underlying similarity graph.

Determination of the Number of Clusters The optimal number of clusters was determined through analysis of the eigenstructure of the graph Laplacian derived from the similarity matrix W . The smallest eigenvalues of the Laplacian and the corresponding differences between consecutive eigenvalues are shown in Figure 6.

Inspection of the eigengap values reveals a distinct separation between successive eigenvalues at $k = 4$. This dominant gap indicates that the similarity graph contains four principal connectivity structures. Beyond this point, the eigenvalues increase more gradually, suggesting that additional components describe internal variation within groups rather than distinct partitions of the graph.

Accordingly, the spectral clustering procedure was applied with four clusters, as indicated by the eigengap heuristic.

Global Distribution of Demographic Trajectory Clusters The spatial distribution of the resulting clusters is presented in Figure 12. Each country is coloured according to the cluster to which it was assigned by the spectral clustering procedure. The map reveals a coherent global organisation of demographic trajectories, with countries exhibiting comparable patterns of age structure evolution grouped together.

While the clusters are not determined by geographical proximity, notable regional consistencies can be observed. Countries within the same region frequently share similar cluster assignments, indicating that the clustering captures geographically

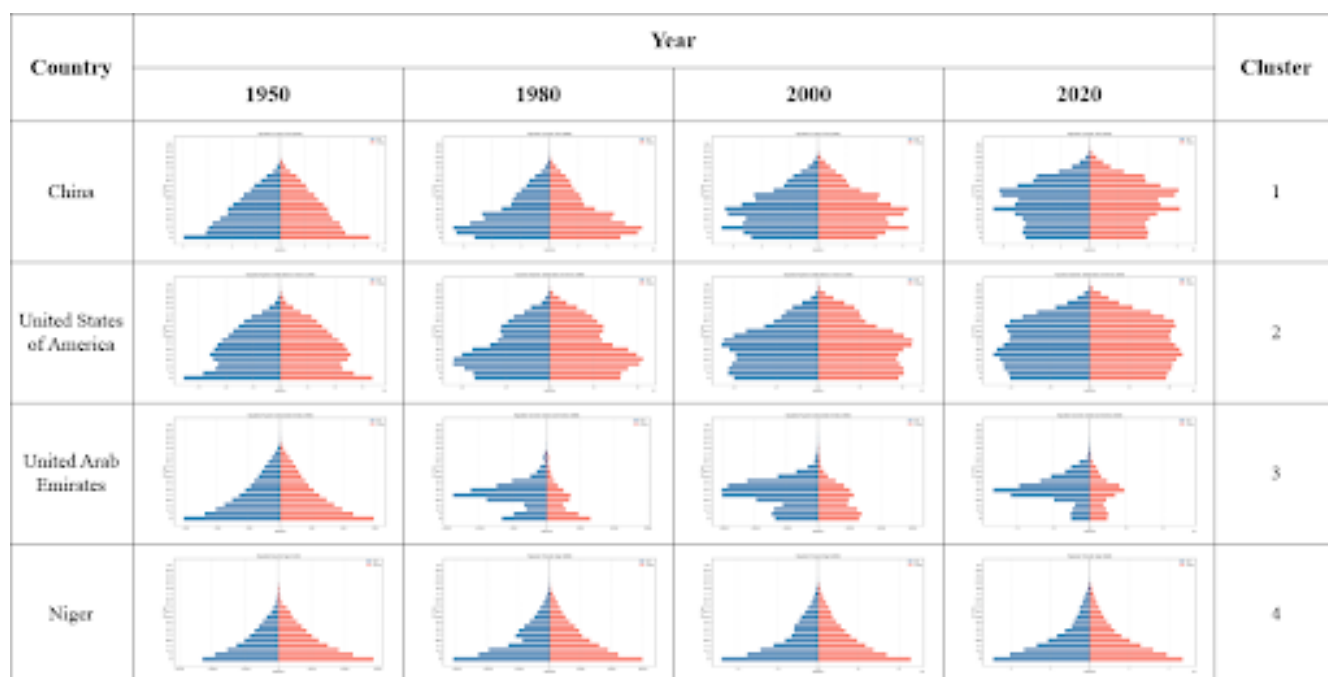


Figure 13. Enter Caption

coherent patterns of population age-structure evolution. At the same time, several clusters extend across multiple continents, indicating that comparable demographic trajectories may arise under diverse geographical and socio economic contexts.

Taken together, these results suggest that the long term evolution of population age structures across countries can be characterised by a limited number of coherent trajectory patterns. The spectral clustering framework therefore provides a useful means of identifying and comparing demographic regimes at the global scale.

Discussion

Global Demographic Evolution

The Global Demographic Evolution framework provides a cross-sectional view of world population structure through time. In this analysis, each decade is treated as a separate demographic snapshot, and countries are grouped according to the similarity of their population pyramids at that particular moment. The purpose is therefore not to trace the full historical pathway of each country, but to examine how the global arrangement of population pyramid regimes changes from one period to another. This allows the analysis to show how demographic regimes emerge, contract, expand, or reorganise across the world between 1950 and 2020.

The results indicate a clear historical reorganisation of global population structure, as shown in Figure 7. In 1950, the world was still largely characterised by youthful demographic regimes, with broad-based and transitional population pyramids accounting for much of the global pattern. Over subsequent decades, this relatively youthful order became increasingly differentiated, as some countries moved towards transitional and ageing structures while others retained strongly youthful pyramids. By 2020, the global demographic landscape had become more mixed, with youthful, transitional, mature ageing, and working-age concentrated regimes coexisting across different parts of the world. Thus, the figure suggests that global demographic change has not produced simple convergence, but rather a plural arrangement of population structures.

The colours shown in Figure 8 correspond to the clusters obtained separately for each decadal time point, but the figure does not by itself assign demographic names to those groups. For interpretation, the clusters were examined after estimation and described according to the average age-structural composition of the countries within each group. Across the study period, these post-clustering descriptions may be organised into several broad regime families: broad-base youthful, youthful transitional, early or late transitional, mature or advanced ageing, and working-age concentrated regimes. These names are therefore not predefined classes imposed on the data, but interpretive summaries of the demographic structures revealed by the clustering procedure. A comprehensive CSV file containing the cluster labels for each analysed year is provided as supplementary material.

The spatial pattern in Figure 8 further clarifies this historical shift. Ageing regimes expand progressively across Europe,

North America, East Asia, and Oceania, indicating the widening geographical presence of mature population structures. At the same time, much of Sub-Saharan Africa remains within youthful regimes, showing that broad-based population pyramids have persisted in a substantial part of the world. Several countries in Asia and Latin America move through transitional regimes, reflecting the gradual narrowing of pyramid bases and the increasing prominence of working-age and older cohorts. A smaller set of countries forms working-age concentrated structures, particularly where migration and labour-market composition have strongly shaped the population pyramid. Thus, the map sequence shows both the global spread of ageing and the continued coexistence of distinct demographic regimes.

These patterns are broadly consistent with the Demographic Transition Model, since the maps show a general movement from youthful population structures towards transitional and ageing regimes^{18,55}. However, the results also refine that model by showing that demographic transition has not occurred uniformly across the world. The timing, pace, and spatial distribution of transition differ substantially across regions, producing a global landscape in which early-transition, late-transition, and ageing regimes coexist. Moreover, the appearance of working-age concentrated clusters indicates that some population structures are shaped not only by fertility and mortality change, but also by migration and labour-market selectivity. The framework therefore extends the Demographic Transition Model by showing how its broad stages are expressed, delayed, accelerated, or altered across different parts of the world.

Nevertheless, this framework has an inherent limitation. Since each analysed year is clustered separately, the approach reveals how the global arrangement of demographic regimes changes at selected time points, but it does not preserve the continuous pathway followed by each country. It can therefore show where countries are positioned within the global demographic landscape at a given moment, but not the full route through which they reached that position. This motivates the country-specific demographic evolution analysis, where each country is treated as a longitudinal trajectory rather than as a series of independent cross-sectional observations.

Country-Specific Demographic Evolution

PC1-Based Country-Specific Demographic Evolution

The objective of implementing the methodological pipeline, which included dimensionality reduction via PCA, functional smoothing through cubic splines, and spectral clustering, was to transform high-dimensional age-cohort data into a biologically meaningful analytical framework. PCA was selected to distill the 21 age variables into an interpretable demographic maturity index (PC1), ensuring that the subsequent reflects the primary axis of variance in global population evolution. Spectral clustering followed by standard k-means was selected as the clustering algorithm to obtain entirely data-driven clusters, avoiding initial assumptions about the number of clusters. Spectral clustering is useful for identifying non-linear similarities within demographic trajectories that traditional Euclidean-based methods may fail to capture. The eigengap curve identified three consistent gaps. As the eigengap heuristic specifically selects partitions that represent stable, persistent structures within the spectral graph, each of the 3 resolutions is inherently reliable and offers complementary layers of information identifying foundational global trends as well as uncovering the detailed demographic journeys at higher resolutions.

PC1 as a Demographic Maturity Index The validation of PC1 as a demographic maturity index is proved by its capacity to map the specific historical and policy-driven demographic trajectories of nations, where a high PC1 value signifies a mature demographic state defined by a mathematical double-positive contribution of a youth deficit and an elderly surplus (Figure 4). Japan is an example for hyper-maturation, with its PC1 score ascending from near zero line in 1950 to an unprecedented peak by 2020. This trajectory has historical roots in its rapid post-WWII transition to world-leading longevity and a sharp fertility decline, resulting in both an extreme youth deficit and an expanding elderly surplus. China's trajectory displays a J-curve, remaining deeply negative during the initial decades before an aggressive breakout after 1970. This reflects the compressed transition associated with state-led fertility restriction and rapid urbanisation^{56,57}. This reflects the compressed transition due to the implementation of the one-child policy and rapid urbanization. The United States represents a relatively linear maturation path, starting the period as an already matured developed society and maintaining a more moderate aging velocity due to a higher fertility cushion, supported by birth rates closer to the replacement level for three decades (1980 to 2007) and the consistent immigration of younger individuals compared to Japan or Europe. This fertility cushion is becoming increasingly fragile as U.S. fertility hit a record low in 2024⁵⁸, putting the US among many other Western nations with low fertility rates. This is a substantial shift for a nation that sustained near-replacement fertility among developed nations. Niger's position as an expansive demographic trajectory with extreme negative PC1 scores provides a baseline for the pre-transitional regime, where high fertility rates persistently regenerate a massive youth surplus. These interpretable trajectories in scores confirm that PC1 effectively distills complex longitudinal structural changes into a single, sociologically valid dimension of maturity.

Interpretability of Functional Motifs The emergence of distinct functional motifs is rooted in the different stages of the demographic transition, a historical process that shifts societies from high to low birth and death rates. Japan and several European nations are at the global extreme of this transition, belonging to the persistent aging ascent, reflecting a terminal stage

of maturation with sub-replacement fertility and record longevity. The persistent youth surplus motif of nations like Niger represents societies still completing the initial phases of demographic transition, where high fertility rates reinforce a young population surplus, preventing the overall age structure from shifting toward older groups. The macro-level bifurcation between China being in the older motif and India in the younger motif at $k=2$ exposes the impact of state-led interventions, such as the one-child policy in China, compared to a gradual demographic evolution in another developing large economy like India, where China's trajectory is consistent with a compressed fertility transition associated with rapid urbanisation and state-led fertility restriction.

Interpretability of Clusters

Policy Interventions Global demographic cluster classifications are noticeably shaped by policy interventions. China's one-child policy and industrialization accelerated its transition into the older population motif by suppressing fertility much faster than its regional peers. Despite Bangladesh and Pakistan sharing a common political history until 1971, they now occupy different motifs at the highest resolution ($k = 5$). Bangladesh's J-curve trajectory remained stagnant at an extreme youth surplus until a sharp breakaway towards maturity after 1970, whereas Pakistan's U-shaped path experienced a younging phase that reached its trough in the mid-1980s before beginning a gradual recovery toward maturity. The disparity is evident in their policy priorities. While Bangladesh's trajectory is consistent with the combined influence of female education, community-level family planning, and sustained fertility decline, Pakistan's policy interventions were limited by fragmented reproductive health services and lower female educational attainment^{59,60}. Albania occupies a distinct position as a demographic outlier in Europe, characterized by its unique placement in the U-rebound towards aging cluster, while its regional peers predominantly occupy more mature post-transitional or persistent aging categories. This is rooted in their radical pronatalist policies, limited access to contraception and abortion, and incentives for large families^{61,62} with 1960 average of seven children per woman. A paradox emerged despite these state interventions, as fertility rates slowly declined due to drivers such as near-universal literacy and high female labor participation. Following the fall of communism in 1991 and the full legalization of abortion from 1995, the fertility rates of Albania plummeted.

Economic Stages Demographic transition showcases alignments with economic development. Transitioning from labor-intensive to capital-intensive economies has often occurred simultaneously with transitioning from young to aging populations. Ethiopia, with a predominantly agrarian-based economy, belongs to the persistent young surplus demographic stage, providing the young manual labor required to sustain the economy. Meanwhile, Vietnam represents the U-rebound towards aging, as global manufacturing and urban industrialization have shifted the demographic curve upward. Brazil in the J-curve ascent cluster demonstrates how mid-century urbanization and shifting toward a service economy are linked to an abrupt drop in surplus of youth. South Korea, with a post-transitional aging ascent, exemplifies how a hyper-competitive, high-tech economic model has compressed the aging process. Japan, with a persistent aging ascent, represents the terminal stage of demographic transition where a wealthy, post-industrial economy comprises an already mature population with a consistently high life expectancy and low replacement, shifting the economic focus from growth to sustainability.

Extra Information Provided in Each Resolution Multiple resolution levels allow discovering patterns ranging from broad global binaries to specific demographic trajectories. For example, the macro resolution ($k = 2$) identifies Sri Lanka as a regional outlier, placing it in the older population motif, while all the other South Asia countries are in the younger population motif. This advanced state of maturation signifies Sri Lanka's early investments in healthcare and education, despite its current lower-middle-income status. At $k = 3$, Sri Lanka is placed in the same cluster with its South Asian neighbors in the transitioning motif, distinguishing it from established aging populations like Japan or Germany. At $k = 5$, Sri Lanka is placed in the young surplus to aging: J-curve ascent, a motif characterized by a sharp acceleration in the Demographic Maturity Index (PC1) since the 1980s, highlighting the extreme velocity of maturation. In 2035, the child percentage (less than 15 years of age) and the elderly percentage (age 70+ years) of the Sri Lankan population are predicted to equilibrate at 21%⁶³. Therefore, the analysis confirms that while Sri Lanka appears older at a macro level, it faces the specific granular challenge of a "J-Curve" transition, where it has the risk of managing a rapidly aging population before achieving the economic resources of already developed or rapidly developing nations.

Multiple-PC Country-Specific Demographic Evolution

Although the first principal component provides a useful and interpretable summary of demographic maturation, it necessarily reduces the evolution of the population pyramid to a single dominant axis. This is valuable for identifying the broad movement from youthful to ageing population structures, but it does not exhaust the information contained in the full age distribution. Population pyramids may differ not only in their general level of ageing, but also in the relative prominence of working-age cohorts, the persistence of youth bulges, the balance between middle and older age groups, and the distortions introduced

by migration or other demographic shocks. For this reason, the analysis was extended beyond the first principal component by retaining the smallest set of components required to preserve the major part of the structural variation in the normalised population pyramids. This allowed each country to be represented not merely by a scalar ageing index, but by a richer low-dimensional trajectory that more faithfully reflects the changing morphology of its population structure over time.

Methodological Basis of Trajectory Comparison Having represented each country by a sequence of principal component vectors, the analysis then treats demographic change as a trajectory rather than as a collection of independent observations. This is important because the transformation of population structure is seldom synchronised across countries. Two countries may pass through broadly similar demographic pathways while doing so at different historical moments or at different rates. A rigid year-by-year comparison would therefore risk understating their substantive resemblance. Dynamic Time Warping was used to address this difficulty by allowing flexible alignment between country trajectories, so that similarity is assessed in terms of the overall form of demographic evolution rather than exact agreement in calendar time. The resulting trajectory dissimilarities were then converted into an affinity structure and examined through spectral clustering, enabling the grouping of countries according to the internal organisation of their long-term demographic pathways.

Evidence for the Four-Cluster Structure The similarity structure obtained from the trajectory comparisons provides important support for the resulting classification. When the affinity matrix is ordered according to the final cluster assignments, it displays a clear block-like organisation, indicating that countries within the same group are more strongly connected to one another than to countries outside that group. This suggests that the observed classification is not merely a numerical convenience, but reflects recurrent patterns in the long-term evolution of population pyramids. The eigengap analysis strengthens this interpretation by indicating that four principal connectivity structures are supported by the graph. Additional divisions would therefore be more likely to describe finer variation within these broad regimes than to identify fundamentally separate demographic pathways.

Interpretation of the Four Demographic Regimes The four clusters may be interpreted as broad demographic regimes defined by the long-term evolution of population pyramid form. One group represents populations that remain strongly youthful, with broad-based pyramids and comparatively limited structural ageing, a pattern most clearly associated with many Sub-Saharan African countries. A second group captures countries that began with youthful age structures but have experienced substantial fertility decline and rapid movement towards a more mature demographic profile, including several emerging economies in Asia, Latin America, North Africa, and the Middle East. A third group represents post-transition ageing societies, where the pyramid base has narrowed and older cohorts have become increasingly prominent, as observed in much of Europe, Japan, North America, Australia, and New Zealand. The remaining group is more exceptional, containing countries whose age structures are strongly shaped by migration and labour-market composition rather than by the conventional fertility and mortality sequence alone. Together, these regimes show that the clustering does not merely arrange countries along a simple young-to-old scale, but separates distinct forms of demographic evolution.

Relation to the Demographic Transition Model These regimes are broadly consistent with the Demographic Transition Model, but they also show why a purely stage-based interpretation is insufficient. The youthful cluster corresponds closely to populations that remain in the earlier phases of transition, where mortality decline has not yet been accompanied by a comparable reduction in fertility. The rapidly ageing emerging-economy cluster reflects countries moving through the later stages of transition, often with considerable speed. The post-transition cluster represents societies in which fertility decline and increasing longevity have already produced sustained population ageing. However, the migration-shaped cluster demonstrates that not all demographic trajectories can be explained by the standard fertility-mortality sequence. In this respect, the results do not simply reproduce the Demographic Transition Model, but refine it by identifying both its principal empirical pathways and the exceptional cases that fall outside its conventional narrative.

Strengths, Limitations, and Implications The principal strength of this approach is that it preserves the temporal and morphological richness of population pyramid evolution while remaining sufficiently compact for comparative analysis. By using more than one principal component, the method retains information that would be lost in a single-index representation; by comparing trajectories, it treats demographic change as a historical process rather than as a sequence of isolated states; and by using the eigengap structure of the similarity graph, it allows the principal grouping pattern to arise from the data itself. Nevertheless, the method also has limitations. The retained dimensionality depends on a chosen variance threshold, Dynamic Time Warping may reduce sensitivity to exact timing, and the final clusters require demographic interpretation rather than mechanical acceptance. Even so, the results provide a useful basis for comparing countries according to long-term demographic pathways, with relevance to health planning, labour markets, pension systems, education, and urban policy. More broadly, the analysis shows that the full population pyramid, when treated as an evolving structure, can serve as a powerful object for studying global demographic transformation.

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Author contributions statement

Must include all authors, identified by initials, for example: A.A. conceived the experiment(s), A.A. and B.A. conducted the experiment(s), C.A. and D.A. analysed the results. All authors reviewed the manuscript.

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